PALEOGEOGRAPHY OF THE NORTH AND SOUTH CHINA BLOCKS DURING THE CAMBRIAN

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ABSTRACT

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Paleomagnetic results demonstrate that during the Cambrian the South China block was close to the equator. We suggest that it was adjacent to North Australia. This reconstruction juxtaposes Cambrian marine basins in South China and Australia, explaining the affinity between Cambrian trilobites from the two areas, as well as the existence of phosphorite deposits in the Early and Middle Cambrian in Australia and in South China. The stratigraphic similarity between the late Precambrian Sinian System in South China and the Adelaide System in Australia, and the continuing fossil affinities from Cambrian through Ordovician of both areas suggest that the proposed geographic configuration lasted from the late Precambrian (800 Ma) to Early Ordovician (470 Ma). Paleomagnetic results from the Cambrian of North China indicate that it was in the southern hemisphere at that time. Based on the paleontological evidence, we suggest that the North China block was close to Tibet, Iran and northern India during the Paleozoic.

INTRODUCTION

Preliminary Apparent Polar Wander (AWP) paths for the North and South China blocks have been obtained (Lin, 1984; Lin *et al.*, 1985). In this paper, we use the paleomagnetic results, in conjunction with paleontological and other geological data, to improve Cambrian paleogeographic reconstructions.

PALEOMAGNETIC RESULTS

Results from the South China Block

Paleomagnetic samples were collected in the summers of 1980, 1981 and 1983, from Zhejiang, Yunnan and Hubei provinces of South China

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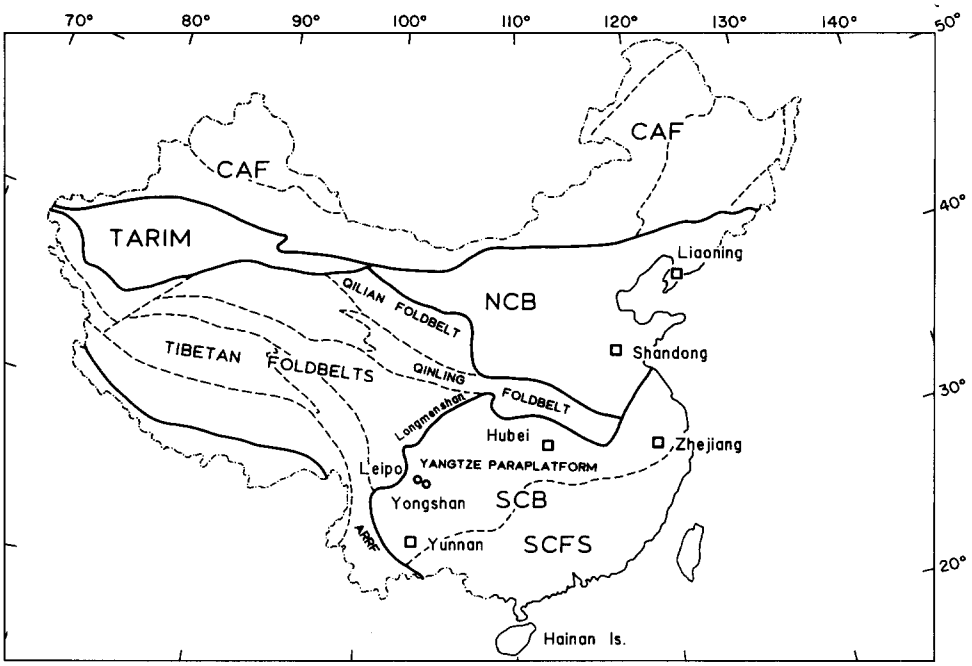


Fig. 1. Map of China showing collection areas in Zhejiang, Hubei and Yunnan provinces of the South China block (SCB), and in Shandong and Liaoning provinces in the North China block (NCB). The SCB consists of the Yangtze Paraplatform and the south China fold system (SCFS). ARRF stands for the Ailaoshan-Red River fault, CAF for the Central Asia foldbelts. The demarcation of the tectonic blocks is based on Huang et al. (1980).

(Figure 1). The sites are all on the Yangtze Paraplatform. However, the term South China block (SCB), as employed in this paper, includes both the Yangtze Paraplatform and the South China fold system of Huang (1978). The SCB is bounded by the Qinling fold belt to the north, the Longmenshan in the northwest, and the Ailaoshan-Red River fault in the west and southwest. Geographically, the SCB includes most of South China, the continental shelf of the South China Sea and the southern part of the East China Sea, northern Viet Nam, and probably the southern part of the Korean Peninsula.

The samples discussed in this paper are from the Hetang Formation in Zhejiang, the Shuijingtuo and the Tianheban formations in Hubei, and the Meishucun Formation in Yunnan. All of these are of Early Cambrian age (Table I). The Meishucun Formation is Tommotian in age with abundant small shelly fossils. The Hetang and Shuijingtuo formations are of the same age. However, the Tianheban Formation is a little younger than the Hetang and Shuijingtuo formations, while the Meishucun Formation is a little older

TABLE I

Stratigraphic Correlation in South China (Sinian – Lower Cambrian)

System	East Yangtze Formation (Member)	Gorges Area Ages (My)	Zhejiang Formation	East Yunnan Formation
Lower Cambrian	Shilongdong		Dachenling	Longwangmiao
	Tianheban △			
				Conglungpu
	Shipai		Hetang △	
		573 ± 07 (Rb-Sr)		
	Shuijingtuo △	574 ± 20 (Rb-Sr)		Qiuzhusi
		568 ± 12 (U -Pb)		
	Tianzhushan			Meishucun △
610 ± 10	Dengying Baimatuo	602 ± 15 (Rb-Sr)		588 ± 13 (Rb-Sr)
			Dengying △	
	Shibantan			Dengying
	Hamojing		(Xifengsi)	
		691 ± 29 (Rb-Sr)		
Sinian	Doushantuo △	700 ± 05 (Rb-Sr)	Doushantuo △	Doushantuo
	Nantuo		Leigongwu △	Nantuo △
	Datangpo			
	Gucheng			
	Liantuo		Zhitang △	Chengjiang △
		740 ± 16 (U -Pb)	(Luojiamen)	
760 ± 20		806 ± 05 (Rb-Sr)		
	Huangling	823 ± 07 (K -Ar)		
Pre-Sinian	Granite	805 (K-Ar, peg)	Baisha Gr.	Kunyang Gr.
		819 (Rb-Sr)		
		860 (U -Pb)		

△ Paleomagnetic Sampling Sites

Stratigrapic correlation of the Sinian-Lower Cambrian formations in eastern Yangtze Gorge area, Zhejiang and East Yunnan of South China. Radiometric ages are from Ma *et al.* (1983). Formations from which we collected paleomagnetic samples are marked with triangles.

than them. These differences in ages should not be significant for the paleogeographic reconstruction, because the rate of apparent polar wander seems to have been low in the latest Precambrian and the earliest Cambrian (Kirschvink, 1978).

The Hetang and Shuijingtuo formations are both black shales. Twenty-one drill samples were collected from 4 sites in the Hetang Formation, and 16 hand samples from 4 sites in the Shuijingtuo Formation. The Tianheban Formation is mainly carbonate, from which 14 samples were collected at 3

sites. The Meishucun Formation is the lowest Cambrian stratigraphic unit and contains a continuous Cambrian/Precambrian boundary section. It is being considered as a candidate for the Cambrian/Precambrian boundary stratotype. It consists of carbonate and siliceous phosphorite. Thirty-nine samples were collected from 13 stratigraphically different levels in the section across the Cambrian/Precambrian boundary.

Remanent magnetization was measured wih a cryogenic magnetometer at the University of California, Santa Barbara. Thermal and AF magnetic cleaning techniques were used to separate the various characteristic components.

The Hetang, Shuijintuo and Tianheban formations show similar behavior during thermal demagnetization. Their natural remanent magnetization is close to the present field direction. However, during thermal cleaning, the magnetization moves away from the present field direction. The intensity decreases to less than 20 percent of its initial value after demagnetization at

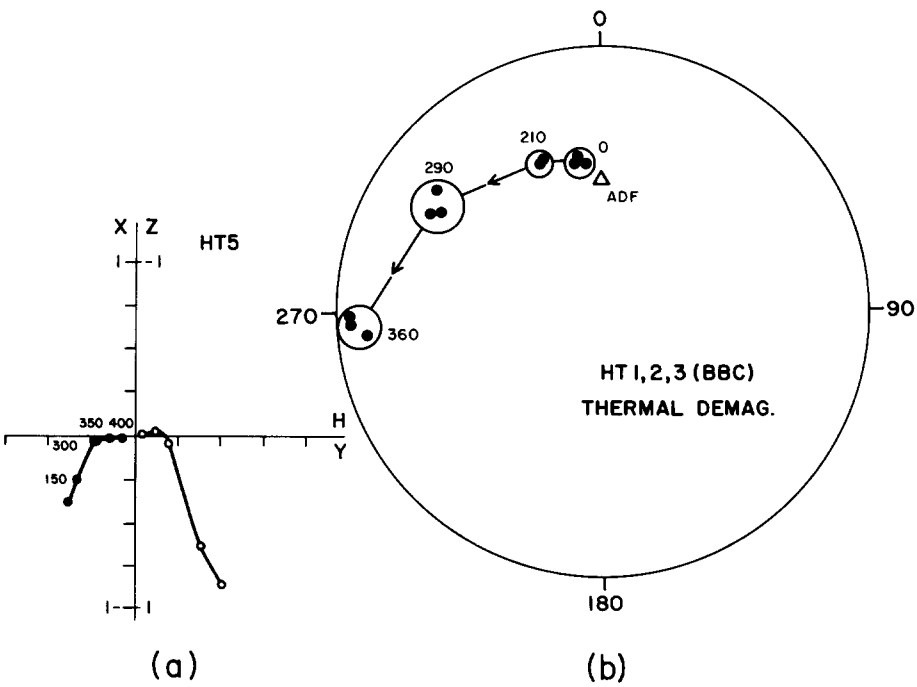


Fig. 2. Thermal demagnetization of samples from the Hetang Formation. (a) Zijderveld diagram of sample HT5. X is to the north, Y to the east and solid circles give data points in this plane. Z is the vertical component and H the horizontal component, and open circles give data points in the vertical plane defined by these two components. Data are plotted after bedding correction (ABC). (b) Equal area projection plots for thermal demagnetization. Solid symbols are for lower hemisphere data. Data are plotted before bedding correction (BBC). ADF is the present Axial Dipole Field in the sampling area.

300 degrees. A single component remains, which is different from the present field direction and from Mesozoic characteristic directions. A single Zijderveld plot is shown in Figure 2a, which illustrates the behavior of these samples. In Figure 2b the changes in magnetization direction brought about by thermal demagnetization in 3 samples from the Hetang Formation are shown.

The Meishucun Formation is difficult to interpret because the AF and thermal spectra of the primary and the secondary magnetizations overlap. A consistent final direction is therefore hard to isolate, although demagnetization produces migration of the magnetization towards the direction defined by the Hetang and Tianheban formations. Figure 3 gives

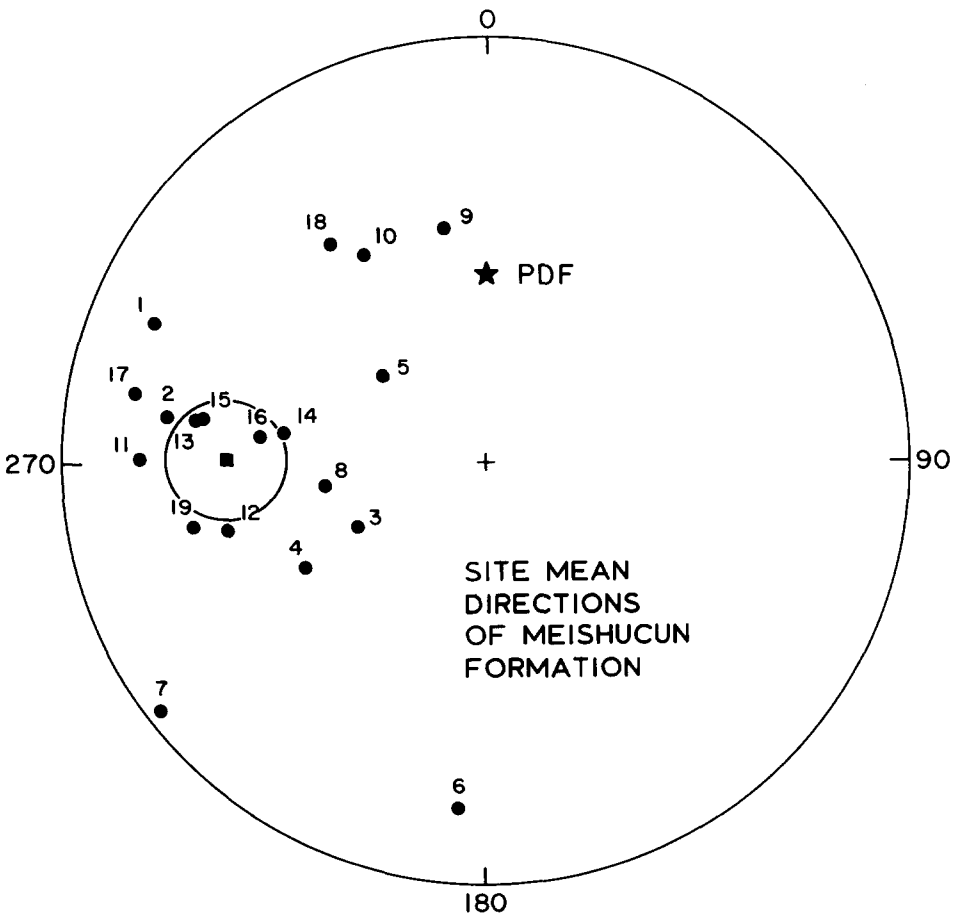


Fig. 3. Stratigraphic level site mean directions of the Meishucun Formation after thermal demagnetization. Solid dots denote the lower hemisphere projection. PDF is the Present Dipole Field in the sampling area. The solid square and the circle are the mean direction and 95% error, respectively, for the 13 sites. Beds are approximately horizontal.

the results for all of the stratigraphic levels collected in the section. The result from each level is based upon three independently oriented cores. As each stratigraphic level is an independent determination of the field, each is regarded as a site. The means for the sites were calculated from the thermally demagnetized data. Sites 5, 9, 18 and 10 were not included in the mean on the grounds that they exhibited present field contamination. Sites 6 and 7 were not included because they appear to be reflecting some other magnetization event. It should be noted that the inclusion of these sites does not significantly alter the mean. The results for all of the South China data are given in Table II.

These characteristic directions, especially those from the Hetang, Tianheban and Meishucun formations, are different from the present field direction. They are also dissimilar from any post-Carboniferous characteristic directions. The collection sites are distributed over distances of several thousand kilometers (Figure 1). Moreover, while the Meishucun and Tianheban formations are almost horizontal, the Hetang Formation dips 60-70°. This provides an opportunity for an attitude test, i.e., to compare their remanent directions before and after bedding correction, which is essentially similar to a fold test. As the relevant sites in the two formations are more than 1000 kilometers away from each other, pole positions, rather than remanent directions, are compared. We used the approach proposed by McFadden and Jones (1981). The hypothesis that the Hetang and Tianheban formations have common true pole position before their bedding corrections can then be rejected at the 95% level. On the other hand, the hypothesis that the two formations provide a common true pole position after their bedding correction cannot be rejected at 95% level. (The critical data are given in Table III.) Thus the Hetang and Tianheban formations pass an attitude test. As the tilting of the Hetang Formation in SE China was from an Early Paleozoic tectonic event, the results from the two formations probably represent the characteristic Early Paleozoic direction.

Results from the North China Block

The Cambrian poles of the NCB are from the Maozhuang and Xuzhuang formations of Shandong and Liaoning provinces, consisting of red oolitic dolomites. After thermal demagnetization to 500 degrees a well defined characteristic direction was obtained (Inc = 55°, Dec = 170°). The bedding is almost horizontal and the direction is quite different from either the present field or directions of any other available Phanerozoic results (Lin, 1984) (Fig. 4). The direction is consistent with an Early Cambrian direction obtained from North Korea (Gurariy *et al.*, 1966). We therefore interpret it as the characteristic Cambrian direction.

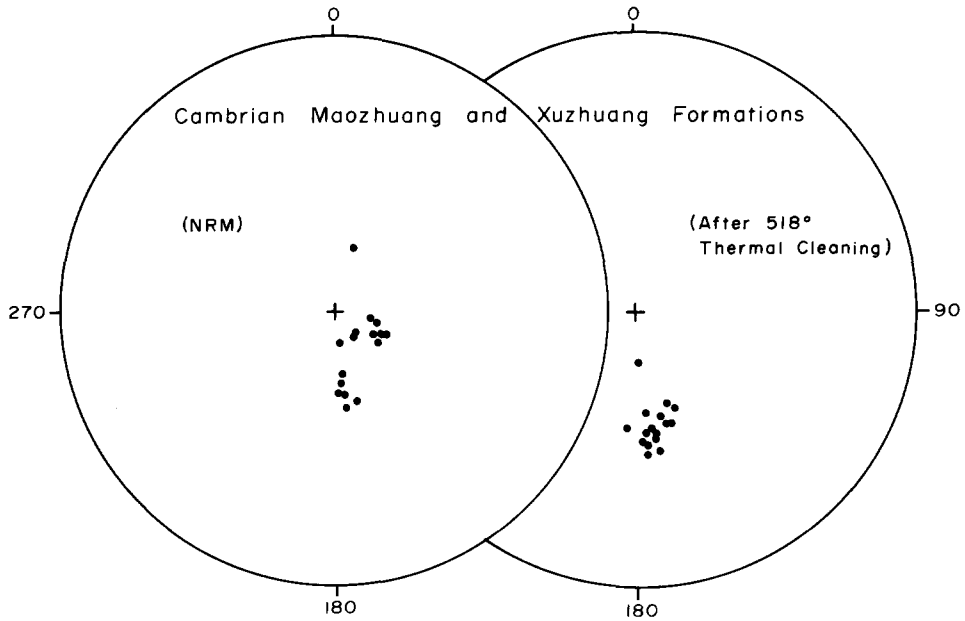


Fig. 4. Magnetization direction from the Cambrian Maozhuang and Xuzhuang formations of the North China block, (a) the Natural Remanent Magnetization (NRM), and (b) directions after thermal cleaning at 518°C. Solid dots are the directions on the lower hemisphere.

Summary

The paleomagnetic poles from the Hetang, Tianheban and Meishucun formationsof the SCB are consistent. However, the pole position from the Shuijingtuo Formation is discordant, probably due to unsuccessful cleaning. The mean of the three consistent results is used to modify the Cambrian World Map of Bambach *et al.* (1980). The mean Cambrian pole position for the NCB is taken from the Maozhuang and Xuzhuang formations. The modified Cambrian World Map is shown in Fig. 5. The major changes are:

1. the NCB and the SCB are treated as separate blocks;
2. the SCB is placed on the equator and adjacent to North Australia instead of at 50°N.;
3. the NCB is placed at 35 degrees in the southern hemisphere close to Iran, Tibet and northern India instead of at 40°N.

The modifications in latitude and orientation are required by the paleomagnetic results from the SCB and NCB. However, the paleomagnetic data alone cannot give a unique reconstruction because of the indeterminacy of paleomeridians. The geologic arguments we used in the

TABLE II

Paleomagnetic Results from Lower Cambrian Formations in South China

1 Number	2 Age	3 Rock Unit & Accessory Information		4 Slat	5 Slong	6 B	7 N	8 TR	9 REV	10 DEC	11 INC	12 KD	13 ED	14 Plat	15 Plong	16 DP	17 DM
SC.064	CBL	Limestone, Tianheban FM, Liantuo, Hubei (Lin)	Site 1	30.8	111.2	1	5	T	0	85.0	29.8	145	6.4	12.3	189.8	3.9	7.1
SC.065	CBL	Same	Site 2	30.8	111.2	1	3	T	0	81.5	13.1	28	23.9	10.7	199.8	12.4	24.4
SC.066	CBL	Same	Site 2B	30.8	111.2	1	6	T	0	103.8	32.5	39	10.8	−2.3	179.1	6.9	12.2
SC.067	CBL	Mean of Tianheban FM				3	14	T	0	89.6	25.5	29	23.2	7.1	189.8	13.5	25.0
SC.068	CBL	Black Dolomite, Hetang FM, Linan, Zhejiang (Lin)	Site 81	30.2	119.7	1	5	T	100	247.6	−13.1	77	8.8	−0.6	201.7	4.6	8.9
SC.069	CBL	Same	Site 31	30.2	119.7	1	5	T	100	272.9	−37.0	28	14.7	7.8	190.8	10.1	17.2
SC.070	CBL	Same	Site 32	30.2	119.7	1	5	T	100	271.1	−42.2	32	13.8	11.1	188.4	10.4	16.9
SC.071	CBL	Same	Site 33	30.2	119.7	1	6	T	100	278.5	−15.8	8	24.4	−3.2	198.9	12.9	25.1
SC.072		Mean of Hetang FM (4 sites)		30.2	119.7	4	21	T	100	274.5	−27.0	29	17.2	3.4	195.6	10.2	18.7

SC.073*u	CBL	Black Shale, Shuijingtuo FM, Liantuo (Lin)	Site 1	30.8	111.2	1	4	T	0	70.3	49.1	67	11.3	30.4	182.3	9.9	15.0
SC.074*u	CBL	Same	Site 2	30.8	111.2	1	4	T	0	72.5	50.1	36	15.6	29.0	180.6	13.9	20.8
SC.075*u	CBL	Same	Site 3	30.8	111.2	1	4	T	0	57.9	53.5	581	3.8	41.7	181.3	3.7	5.3
SC.076*u	CBL	Same	Site 4	30.8	111.2	1	4	T	0	76.9	52.4	54	12.6	26.2	176.8	11.9	17.3
SC.077*u	CBL	Mean of Shuijingtuo FM				4	16	T	0	69.5	51.5	227	6.1	31.8	180.2	5.7	8.3
SC.078	CBL	Limestone, Meishucun FM, Jinning, Yunnan (Lin)		24.4	102.3	13	39	T	100	271.1	37.5	17	10.2	9.4	211.1	7.1	12.0
	CBL	Mean of Hubei, Zhejiang & Yunnan (Tianheban, Hetang & Meishucun FMS)				20	74	T	85			28	8.8	3.4	195.0		

Paleomagnetic results from Lower Cambrian formations. Column 1 shows the pole numbers on the China Paleomagnetic List (Lin, 1984). Column 2 denotes the age of the poles; here CBL stands for Lower Cambrian. Column 3 provides information about the rock unit, rock type, site, and authors. Column 4 and 5 give the latitude and longitude of the sampling site. Column 6 and 7 give the number of sites (B) and the number of samples (N). Column 8 denotes the cleaning techniques used, T for thermal, A for AF demagnetization. Column 9 gives the proportion of the reversed samples (reversed over the total samples). Column 10 and 11 give the remanent declination and inclination. Column 12 (KD) and 13 (ED) give estimates of the precision parameter (K) and the confidence interval at the 95% level ($\alpha 95$ or A95). Column 14 and 15 give the latitude and longitude of paleomagnetic pole position. Column 16 and 17 give the semi-axes of confidence ellipse of pole position, with DP being along the paleolongitude which passes through the sampling site, and DM perpendicular to DP.

TABLE III

Results for the Hetang and Tianheban formations in an attitude test.

	Before Bedding Correction (BBC)		After Bedding Correction (ABC)	
	Pole Position		Pole Position	
Site	Lat.	Long.	Lat.	Long.
(1) Hetang Formation in Zhejiang Province				
81	−4.7	212.8	−0.6	201.7
31	9.3	211.6	7.8	190.3
32	12.3	211.3	11.1	187.8
33	−11.5	212.5	−3.2	198.5
	RB (HT) = 3.941285		RA (HT) = 3.959427	
	KB (HT) = 51.09		KA (HT) = 73.94	
(2) Tianheban Formation in Hubei Province				
1	8.8	188.0	12.3	189.8
2	3.8	196.7	10.7	199.8
2B	−6.3	178.6	−2.3	179.0
	RB (TH) = 2.957212		RA (TH) = 2.948157	
	KB (TH) = 46.74		KA (TH) = 38.58	

Overall statistics

BBC: Rb = 6.746538 Kb = 23.67 N = 7

ABC: Ra = 6.898225 Ka = 58.95 N = 7

modification are discussed in the following sections. We use the term 'geosyncline' because it affords a continuity with the older literature. However, we do not wish to imply by its use the generic meaning the term had in the past.

THE CAMBRIAN MARINE BASINS

Figure 6 shows the restored configuration between Australia and the SCB during Early Cambrian. It is an enlargement of a part of Fig. 5. According to Brown *et al.* (1968), the Australian Cambrian marine deposits are constrained along a northwest-southeast central belt, including the Tasman Orthogeosyncline, Kanmantoo Trough, Adelaide Geosyncline, Amadeus Basin, Georgina Basin, Ord Basin, Daly River Basin and Bonaparte Gulf Basin (Fig. 7).

In the Early Cambrian, basalts covered the northwest quadrant of the Australian continent, in the Ord and Daly River Basins, whereas terrigenous and carbonate deposits accumulated in the Adelaide Geosyncline and Amadeus Basin, and volcanics, greywackes, and shale in the Tasman Orthogeosyncline. By the Middle Cambrian shallow-water sedimentation had extended to much of the Ord, Daly River and Georgina

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RB and RA are the resultant lengths of the site mean unit vectors before and after the bedding correction, respectively. KB and KA are the precision parameters of the site mean directions before and after bedding corrections, respectively. The Hetang Formation (HT) consists of four sites (81, 31, 32,33), the Tianheban Formation (TH) of three sites (1, 2, 2B). The total number of sites of the two formations is thus $N = 7$. Rb and Ra are the resultant lengths of the site mean unit vectors from both HT and TH formations before and after bedding correction, respectively,
or $|\bar{R}| = |\bar{R}\,(HT) + \bar{R}\,(TH)|$. Based on McFadden and Jones (1981),

$$f(R) = \frac{R\,(HT) + R\,(TH) - R^2/[R\,(HT) + R\,(TH)]}{2\,[N - R\,(HT) - R\,(TH)]}$$

$$f(P) = \left(\frac{1}{P}\right)^{1/(N-2)} - 1$$

When $P = 0.05$ and $N = 7$, we have $f(P) = 0.8206$. Before bedding correction $f(R) = 1.4806$; after bedding correction $f(R) = 0.1012$. As $f(R) > f(P)$ before bedding correction, therefore the hypothesis that the Hetang and Tianheban formations have common true pole position can be rejected at 95% level; on the other hand, as $f(R) < f(P)$ after bedding correction, thus the hypothesis that the two formations provide a common true pole position cannot be rejected at 95% level. At last,

$$\frac{KB\,(HT)}{KB\,(TH)} = 1.0931 \qquad \frac{KA\,(HT)}{KA\,(TH)} = 1.9165$$

both are smaller than $F_{4,6} = 4.53$ in F-test, which indicates that the precision parameters of the two formations are not significantly different both before and after bedding correction. This justifies the procedure used above. In conclusion, the results from the Hetang and Tianheban formations pass an attitude test (or a 'fold' test) at 95% level.

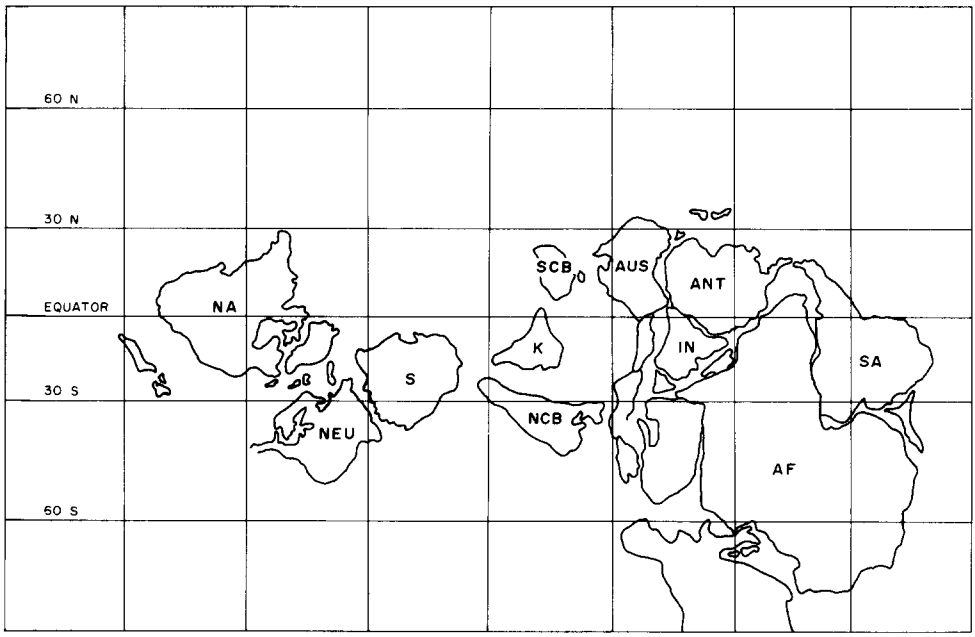


Fig. 5. Early to Middle Cambrian world map. The abbreviations are: AF-Africa, ANT-Antarctica, AUS-Australia, IN-India, K-Kazakhstan, NA-North America, NCB-North China block, NEU-Northern Europe, S-Siberia, SA-South America, SCB-South China block. Note that the North China block and South China block are separated.

Basins; in the Tasman Orthogeosyncline and Amadeus Basin sedimentation continued as in Early Cambrian; but the Adelaide Geosyncline was deformed and sedimentation ceased. In the Late Cambrian sedimentation covered the same areas as in the Middle Cambrian except in the Daly River Basin which became land.

The Cambrian was a quiet period tectonically in Australia except in the Tasman Orthogeosyncline and in the Adelaide Geosyncline. The folding of the Adelaide Geosyncline produced minor deformation in the west, but in the east there was strong overfolding and thrusting. This movement began in the Middle Cambrian, but probably extended into the Ordovician. Elsewhere in Australia the only Cambrian movements were epeirogenic.

According to Wang and Liu (1980), Huang *et al.* (1980) and CGGMA (1982), Cambrian marine sediments covered almost the whole of the SCB. In the far west the Kangdian upland shed material to the east. Therefore, in the South China Cambrian marine basin we see the following pattern of sedimentary facies belts: clastic and argillaceous rocks dominate in the west; in the central region the section consists mainly of argillaceous mudstone

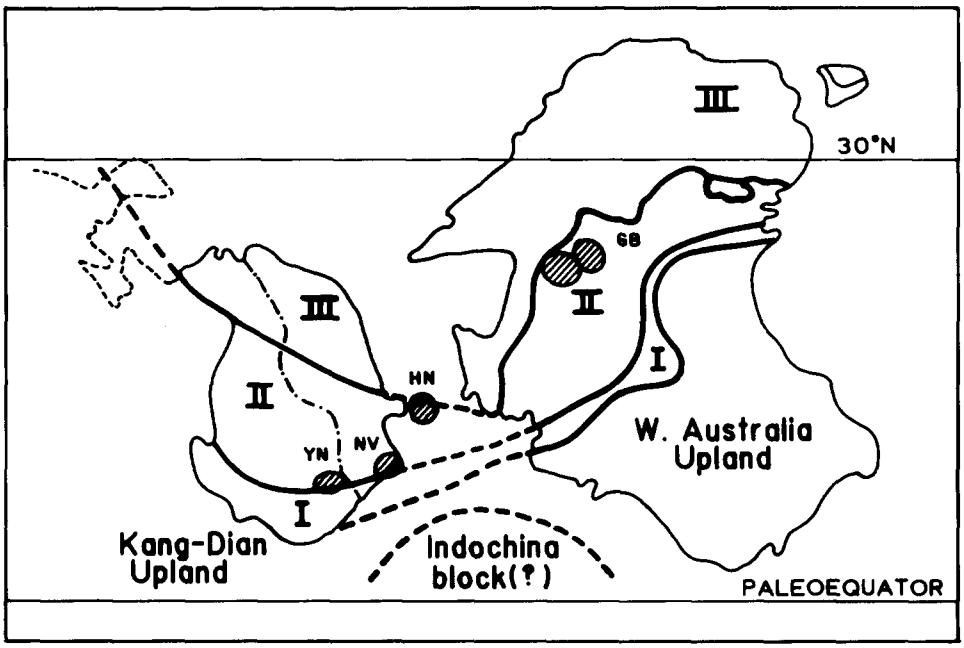


Fig. 6. Juxtaposition of the SCB and Australia in the Cambrian. The sedimentary facies belts denoted are as follows:

I – inner detrital belts (sandstone, shale);

II – intermediate carbonate belt ('miogeosyncline');

III – outer detrital belt (flysch, 'eugeosyncline').

The boundary between the Yangtze Paraplatform and the South China fold system (Huang, 1978) is indicated with a broken line. Southern Korea was probably a part of the South China block. The position of the Indochina block is speculative. Five large Cambrian phosphorite deposits are indicated with the following abbreviations: YN-Yunnan, NV-Northern Viet Nam, HN-Hainan Island, GB-Georgina Basin.

and shale; in the east mainly flyschoid deposits are found. In the southeast, there is the so-called Southeast China 'Caledonian' foldbelt.

The continental shelf in the northern part of the South China Sea between NW Australia and the SCB (Fig. 5), also has a Precambrian crystalline basement. A drill sample from near the Xisha Islands (17N, 112E) yielded an age of 627 Ma from the Rb-Sr whole rock isochron. Further to the west, the Indochina block also has a Precambrian crystalline basement which is overlain by the flat-lying quartzite of Middle Cambrian age (Huang, 1980). It appears that during the Cambrian the west Australian upland may have shed its material eastwards to the central Australian marine basin, and northwards and northwestwards to the marine basins of the Indochina block and the SCB. The directions here are referred to in present geographic coordinates.

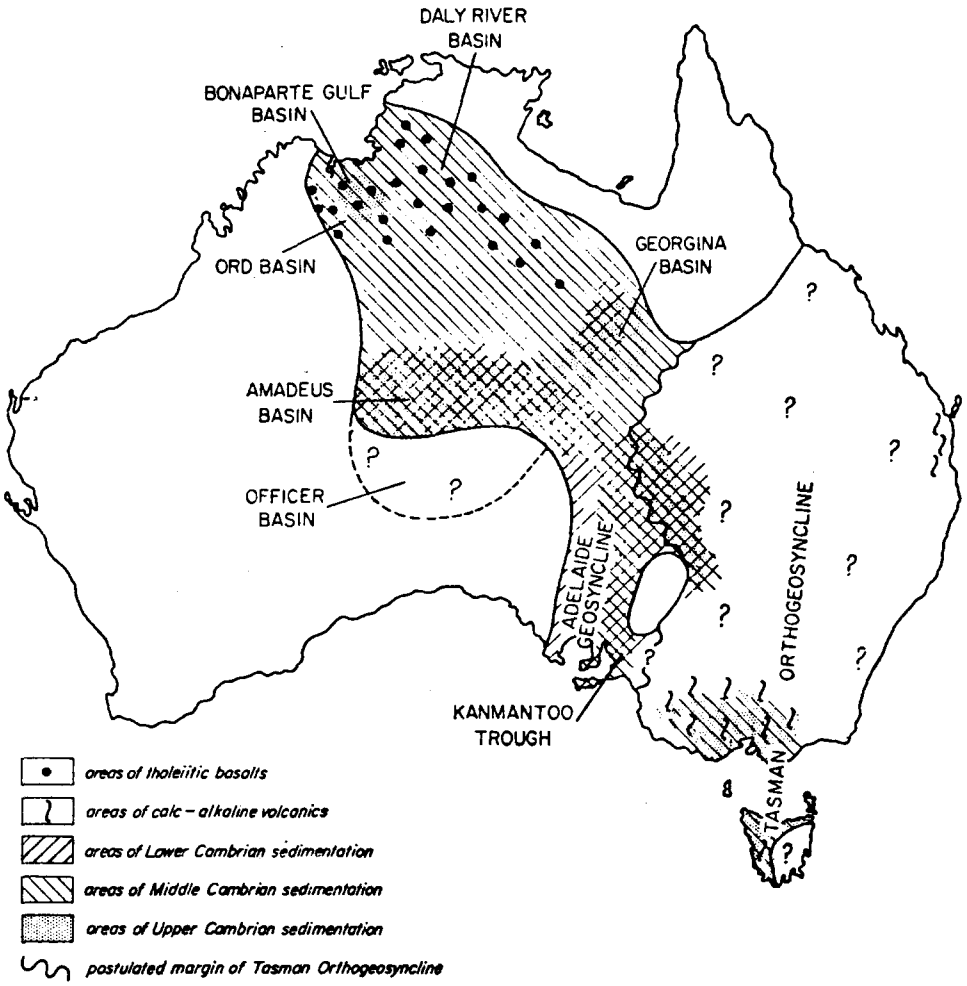


Fig. 7. Cambrian paleogeography of Australia (from Brown et al., 1968).

Palmer (1974) pointed out that three distinctive sedimentary facies belts surround the Cambrian uplands of North America and Siberia. These are an inner detrital belt (marine sandstone, shale from the central land mass), an intermediate carbonate belt (clean limestone and dolomite in shallow seas), and an outer detrital belt (dark-colored thin-bedded siltstone, silty limestone with occasional black cherts indicating a deeper marine facies situated seaward of the carbonate belt).

To the east of the West Australian upland, there are Cambrian marine basins with an inner detrital belt and farther east an intermediate carbonate belt. To the north and northwest such belts are missing in today's

geographic configuration, but in the SCB and the Indochina block just such Cambrian marine basins are found. The quartzite in the Indochina block and the carbonaceous shale and argillaceous rocks in the South China fold system require a Cambrian upland to their south. However, the necessary Cambrian upland is missing in today's configuration.

In the proposed reconstruction, the Indochina block and the South China fold system are brought next to North Australia in such a way that the missing sedimentary basins of Australia are seen to be the South China fold system and the Indochina block. Moreover the sediment source for these basins is seen to be the West Australian upland. The sedimentary facies belts are also in the correct configuration – quartzite typical of the inner detrital belt is found in the Indochina block, and carbonaceous shale and argillaceous rocks typical of the intermediate belt and outer detrital belt are found in the South China fold system. To the southeast, the early Paleozoic Southeast China foldbelt is juxtaposed with the Tasman Orthogeosyncline along the southeastern margins of two continents. This restored pattern of sedimentary facies belts of Australia, the SCB and the Indochina block is similar to those which are seen around the North American Cambrian upland and the Siberian Cambrian upland.

THE CAMBRIAN TRILOBITES

The very strong affinities of Cambrian trilobites in the SCB, Indochina and Australia have long been recognized (Palmer, 1973; 1974; Lu *et al.*, 1974; Lu, 1981). The trilobites can be divided into two categories: endemic, which are confined within a local area, and cosmopolitan, which are widely or world-wide distributed. Based on work in North America, Palmer (1974) noted that the inner detrital and carbonate belts were characterized by shallow-water restricted marine environments containing predominantly endemic trilobites, whereas the outer margin of the carbonate belt and the outer detrital belt were characterized by deep-water environments containing both endemic trilobites and a large number of cosmopolitan species.

Many of the trilobites of Australia, the SCB and Indochina, which show strong affinities to each other, are not only of the cosmopolitan type but also of the endemic; they occur in the carbonate belt and inner detrital belt characterized by shallow-water restricted marine environments. It has long been a puzzle why the endemic fossils in two such widely separated areas are so similar. The proposed reconstruction explains the puzzle because the areas were parts of the sedimentary basins around NW Australia during Cambrian time. They were either adjacent to, or even directly linked to each other, so they shared the same trilobites.

PALEOCLIMATE

The paleomagnetic results from Australia and the SCB indicate that both were at low or intermediate latitude in the Cambrian. This is generally consistent with the paleoclimatic indicators.

In the SCB, Early Cambrian stone-coal (burnable black shale) has been found in Zhejiang, Jiangxi and Guangdong provinces. This stone-coal was formed by algae and indicates the existence in this region of a flourishing algal community, such as might be expected in tropical, or subtropical, environments. The occurence of gypsum and salt pseudomorphs during the Middle Cambrian in Leipo and Yongshan of the border area of Sichuan and Yunnan provinces, and of other evaporites during the Late Cambrian in western Hupei, both of the SCB, indicates that the SCB experienced a hot and arid climate in the Middle and Late Cambrian (Wang and Liu, 1980).

The occurence of salt pseudomorphs and gypsum in Middle Cambrian sediments from the Ord, Daly River and Amadeus Basins, the thick redbed deposits in the Middle Cambrian in the north of the Adelaide Geosyncline, indicate that Australia had warm to hot conditions and widespread aridity in the Cambrian (Brown *et al.*, 1968).

Archaeocyathids have been cited as warm-water indicators. For example, the Cambrian archaeocyathid reef-bearing belts on the Siberian Shield were on or aligned with the paleoequator (Kirschvink and Rozanov, 1983). Archaeocyathids occurred in both Australia and the SCB during the Cambrian, when the latter was equatorial and the former at low paleolatitude.

In summary, the paleoclimatic indicators are consistent with the paleolatitudes suggested by paleomagnetism-based reconstruction.

PHOSPHORITE DEPOSITS

Based on the study of the spatial and temporal distribution of sedimentary phosphorite deposits, Cook and McElhinny (1979) recognized that there are six main periods of phosphorite production; they are Cambrian, Ordovician, Upper Carboniferous to Permian, Jurassic to Lower Cretaceous, Upper Cretaceous to Eocene, Miocene to Recent. Four of them show that the majority of phosphorite deposits formed at low latitudes. The two exceptations with intermediate latitude (30-50 degrees) are the Jurassic to Lower Cretaceous and the Cambrian. The former is mainly due to the contribution from the deposits of central Russia, and the latter is due to the contribution from the deposits of South China.

The proposed intermediate latitude of 40-50 degrees for the Chinese

phosphorite by Cook and McElhinny was deduced from a Cambrian pole from North Korea. Our paleomagnetic results suggest that North Korea has been a part of the North China block (the NCB) since the Cambrian, and that the NCB and SCB were separated from each other during the Cambrian. Therefore the paleolatitude proposed by Cook and McElhinny for the SCB cannot be justified anymore. Our paleomagnetic result shows that the SCB was near the equator during the Early Cambrian. Yunnan Province in SW China, where the major phosphorite mines occur, was at about 10 degrees. When we replot the paleolatitude histogram based on direct measurement, the Cambrian no longer appears to be an anomalous period for phosphorites. The majority of the Cambrian phosphorites were also formed at low latitude (Fig. 8).

Our reconstruction brings close together the four largest Cambrian phosphorite mines in the SCB, North Viet Nam and Australia, each of which is estimated to contain more than 100 million tons (Notholt, 1979). Because they were distributed in the same or adjacent Cambrian basins at low paleolatitude, their formation can easily be interpreted in terms of the equatorial upwelling mechanism. In particular, the phosphorite in the Georgina basin of Australia and in the Hainan Island of the SBC are very similar to each other. For example, (1) both were formed in the Middle Cambrian; (2) both are in similar sedimentary sequences, i.e., from the bot-

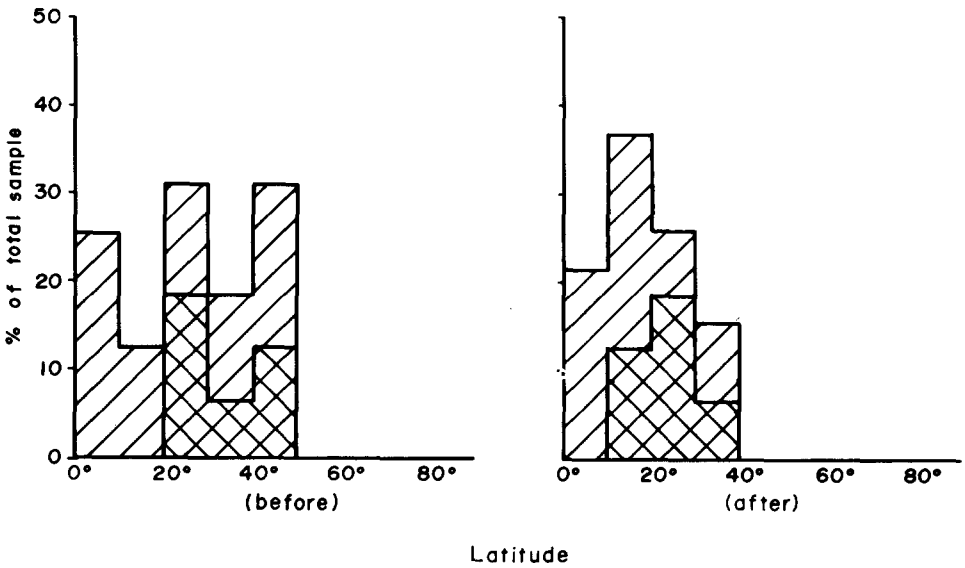


Fig. 8. Paleolatitude distribution of the Cambrian phosphorites. On the left is the original plot (Cook and McElhinny, 1979). On the right is a new plot in which the Chinese phosphorites are assigned the paleomagnetically-determined paleolatitude. The cross-hatched areas represent major phosphorite deposition.

tom, sandstone – siliceous rocks – phosphorite – shale and carbonate; (3) both are strata-type; (4) both are siliceous or biogenetic-siliceous in which the biodetritals are phosphorized with silica as cement; (5) the phosphorized detritals in both cases are either lobate or tubular with the sizes ranging from 0.5 to 1.0 mm for the Hainan phosphorite, and from 0.2 to 0.6 mm for the Georgina phosphorite; (6) both phosphorites evidently formed in a similar environment of warm, shallow-sea and high energy; and (7) both also contain similar fossil assemblages (Lu, 1979). This implies that Hainan Island was close to the Georgina Basin in Middle Cambrian, as now shown in our reconstruction (Fig. 5 and Fig. 6).

The preliminary paleomagnetic results from Kazakhstan show that it was near the equator during the Cambrian. We suggest that the Kazakhstan block was close to the SCB at that time as shown in Fig. 5, because its trilobites show affinities to those in the SCB, Indochina and Australia, and it is also rich in Cambrian phosphorite deposits.

In a recent paper on Cambrian phosphorites, Parrish *et al.* (1983) suggest that the Chinese Cambrian phosphorite might have been formed due to the coastal upwelling on the *west coast* of continents at *mid-latitude.* The suggestion is speculative because of the lack of paleomagnetic data from the Chinese blocks, as they admitted in their paper. Our paleomagnetic results show that: (1) the Cambrian phosphorite in the SCB was formed at an equatorial latitude, not at mid-latitudes; (2) the declination of the Cambrian remanent magnetization is about 90°, so the Cambrian sedimentary facies belts were originally formed parallel to the paleolatitudinal lines (east-west) even though presently they look parallel to meridians. The Chinese paleomagnetic results therefore suggest an equatorial upwelling mechanism for their deposition.

COMPARISON OF THE SINIAN SYSTEM AND THE ADELAIDE SYSTEM

Late Precambrian formations are widely distributed in both the SCB and Australia (Liu *et al.*, 1973; McWilliams and McElhinny, 1980; Ma *et al.*, 1983). Stratigraphically, they are surprisingly similar. Table IV gives stratigraphic sections for the Sinian System in South China, and the middle and upper Adelaide System in Central Australia.

1. The Burra Group (Australia) and the Liantuo Formation (SCB) are similar. Below them there is a significant unconformity in both areas. Below this unconformity, there is the Huangling granite in the SCB which is dated at about 800-860 Ma, whereas in Australia there are the basaltic and trachytic lavas of the Callana Beds which are 867 ± 32 Ma.

2. The erosional surface between the Nantuo and the Liantuo formations in the SCB has its equivalent in Australia, which is the unconformity between the Umberatana and Burra groups.
3. The occurence of the Nantuo tillite formation and the Yudnamutana glacial sequence correlate. The age of the former is 700-740 Ma (Ma *et al.*, 1983) and of the latter 740 Ma (McWilliams and McElhinny, 1980).
4. Above the lower tillite, in the eastern part of the Adelaide Geosyncline, there is the Farina Subgroup consisting of siltstone and shale, and then another tillite formation, the Yerelina Subgroup. In the southeastern part of the SCB, e.g., in SE Guizhou, SW Hunan, and N Guangxi provinces, the lower tillite is followed by sandstone and black shale, and then by the upper tillite; in the western part of the Adelaide Geosyncline there is only one tillite formation. In the western of South China, e.g., in Yunnan, Hupei and N Guizhou, there is also only one tillite formation in the sections.
5. The Doushantuo and Dengying formations in the Sinian Systems and the Wilpena Group in the Adelaide System were both formed in a hot-warm and arid climate.

At the top of the Wilpena Group in Australia is the Pound Quartzite, which contains the well-known Ediacara fauna. There has also been some indication of the Ediacara fauna in South China (Yan and Qian, 11983). However, the poor preservation has made identification difficult, resulting in an argument about its validity. Nevertheless, the Meishucun Formation in Yunnan Province is well-known for its abundance of primitive shelly fossils near the Cambrian/Precambrian boundary, which has recently been suggested to be about 610 Ma in age (Ma *et al.*, 1983).

THE CAMBRIAN POSITION OF THE NORTH CHINA BLOCK

The suggested southern hemisphere position for the NCB might seem to be strange, so that further explanation is necessary. In the NCB there is a great sedimentary hiatus from Upper Ordovician to Lower Carboniferous. The oldest post-Cambrian pole with known polarity is the Permian pole (McElhinny *et al.*, 1981; Lin, 1984). If we assume that the apparent polar wander was less than 90 degrees from the Cambrian to the Permian, which seems reasonable in the light of the APW paths for the other continents, then the polarity ambiguity of the Cambrian pole can be resolved. Based on this suggestion, we propose that the pole position at 15.0N, 298.6E is the

Cambrian north pole. If this is correct, then it follows the NCB must have been in the southern himisphere in the Cambrian.

Although the southern hemisphere position of the NCB in the Cambrian is based on the paleomagnetic results, it is consistent with sedimentary facies belts and fossil records. For example, the NCB in our Cambrian reconstruction is placed close to Tibet, northern India and Iran. This is consistent with the similarity of the trilobites in the NCB and those in Spiti of the Himalayas, in the Great Salt Range of Pakistan, and in Iran (Wang and Liu, 1980). The Cambrian sediments in these areas are of epicontinental or platform-type, e.g., sandstone, shale and carbonate, as they are in the NCB. Moreover, Sonnenfeld (1981) has drawn attention to the nearly continuous presence of time-equivalent evaporite sequences from SW Arabia to NW China (Tarim) in the Cambrian; for example, there are thick Cambrian halite deposits in the Salt Ranges of Pakistan, Rajasthan of India, and Central Afghanistan, and in a belt from Central Iran to the Persian Gulf, and in the Arabian Peninsula. This evaporite facies is correlative with the salt-bearing formation mapped in the Tarim Basin. It was suggested that this Cambrian-Ordovician arid belt originated at a paleoatitude of 30-40°, which is in accord with our paleomagnetic results.

The proximity of Tibet, Northern India and the NCB, as we propose here, may not have been limited to the Cambrian, but probably extended into the middle or even late Paleozoic. For example, (a) based on the study of Chen (1981), Early Ordovician cephalopods are devided into an actinoceroid fauna of the North China type and an orthoceroid fauna of the South China type. In Tibet, actinoceroids make up to 70% of the total amount of the Early Ordovician fossils, attesting its strong affinity to the North China fauna. (b) the sedimentary gap between the Ordovician and the Permian in Iran, Arabia, India and the Tarim Basin is also seen in the NCB, although in the latter case the great hiatus is from the Middle Ordovician to the Early Carboniferous. (c) high-alumina clays in the Permian beds of Iran and in the Upper Carboniferous beds of the NCB both attest to a tropical latitude in the late Paleozoic, which is also consistent with the paleomagnetic results. (d) the occurrence of the Cathaysian flora elements in Iran, Turkey and Iraq also argues that in the late Palezoic those areas, which now form Asia Minor, were not very far from the Chinese blocks

We have suggested elsewhere (Lin *et al.*, 1985) that the NCB was separated from the Asia Minor block(s) in the Late Permian. It was then translated to the northeast with respect to the Angara block, and finally accreted to the Siberian block.

HOW LONG DID THE SUGGESTED PALEOGEOGRAPHIC CONFIGURATION LAST?

The fossil affinity between the SCB and Australia in the Cambrian continued at least until the Ordovician. For example, Whittington (1973) points out that there are four trilobite provinces in the Ordovician: the bathyurid fauna, the asaphid fauna, the Selenopeltis fauna and the hungaiid-calymenid fauna. Both the SCB and Australia belong to hungaiid-calymenid province, together with Himalaya and Pamir. This, together with stratigraphic similarity between the Sinian System and Adelaide System, leads us to suggest that the reconstruction proposed in this paper may have lasted from 800 to 460 Ma.

On the other hand, early Devonian brachiopods may reveal when the SCB separated from Australia. In the early Emsian (Early Devonian) there were apparently three faunal realms based on the distribution patterns of brachiopods. These are the Old World Realm consisting of Europe, Asia, Australia, North Africa and western North America, the Eastern America Realm consisting of eastern North America and also Brazil, Colombia and Venezuela, and the Malvino-Kaffric Realm which includes the southern tips of South America and Africa (Boucot *et al.*, 1969).

The Yukiangian stage in South China is equivalent to the early Emsian. The Yukiangian formations are found in Guangxi, Yunnan, Sichuan provinces and in North Viet Nam. They consist of argillaceous and marly rocks. The brachiopod fossils show high diversity and provinciality. For example, among 35 genera found in the Yukiangian formation, 13 are cosmopolitan types (37 percent), 21 are the Old World types (60 percent), and at least 10 are endemic (about 30 percent). So the Yukiangian formations may well be considered as a special province – the Youngjiang-Longmenshan province under the Old World Realm (Wang and Rong, 1983). In addition to the endemic genera, the Yongjiang-Longmenshan provine is also characterized by: 1. some common genera in West Europe and North Asia have not been found in the SCB, e.g., *Atrypinella*, *Karpinskia*, *Carinatina*, *Carinatinella*, *Gorgostrophia*, *Rhitistrophia*, *Rugoleptaena*, *Ivdelinia*, *Clorinda*, *Clorindina*, *Altajella*, *Nymphorhynchia*, *Glossinotoechia*, *Sibirirhynchia*, *Uralospirifer*, *Costispirifer*; 2. some common fossils of Early Devonian age in the Old World Realm have not been found in the SCB, e.g., *Eospirifer*, *Striisperifer*, *Havlicekia*, *Janius* and so on; 3. among the common genera in Australia and New Zealand, e.g., *Maoristrophia*, *Australirhynchia*, *Planicardinia*, *Sturtella*, *Nadiastrophia*, *Reeftonia*, only *Nadiastrophia* is found in China (Wang and Rong, 1983). The provinciality of the SCB in the Early Devonian may therefore suggest that the SCB had already separated from Australia by the Early Devonian.

The tectonic evolution between the SCB and Australia is inversely related

to that in the North Atlantic. From the late Precambrian to the close of Middle Ordovician, an open ocean existed in the North Atlantic (Wilson, 1966) whereas the SCB and Australia were close to each other, with no wide ocean between them. From Upper Ordovician to the Carboniferous, the proto-Atlantic Ocean was closed (Wilson, 1966), whereas the SCB drifted away from Australia. Thus, the separation between the SCB and Australia in the late Paleozoic appear to correlate with the closing of the proto-Atlantic. This is in contrast to the geological history since the Mesozoic in that the Atlantic has been expanding at the expense of the Pacific for the past 200 Ma. The Cambrian reconstruction and the evolution process presented in this paper, when combined with those for the North Atlantic, provide a global picture of the plate tectonics for the late Precambrian – Early Paleozoic.

The suggestion that there is an inverse or complementary relation between a proto-Pacific and proto-Atlantic can be tested by comparing the APW paths during the time interval. Unfortunately the limited results from both regions make a decisive test impossible at present. However, a rough comparison shows that the suggestion is plausible. For example, the Ordovician, Cambrian and the uppermost Precambrian poles from the South China block are generally limited to the area close to the Phoenix Islands in the equatorial central Pacific. This is consistent with Briden's oldest quasi-static interval from the latest Precambrian to the Ordovician in the Gondwana APW path (McElhinny, 1973; Fig. 136a). Durig the following drift episode from Late Ordovician to Early Devonian the south pole for Gondwana moved from NW Africa to SE Africa whereas the north pole for the SCB moved northeastwards, i.e., they show the same sense of polar wandering during that period. This comparison seems promising although it can only be approximate because of the preliminary nature of the results.

CONCLUSIONS

In summary, paleomagnetic results from the North and South China blocks lead to a reconstruction, placing the SCB next to NW Australia, and the NCB alongside Tibet, Iran and northern India. This is different from previous reconstructions. It explains the strong similarities between the late Precaambrian stratigraphy in the two regions and the affinities between their Cambrian and Ordovician fossils. It also explains the distribution of the sedimentary facies belts, and the occurrence of phosphorite deposits and evaporites.

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